

A FINANCE & ENGINEERING BRIEFING

From Token Bills to Decision Economics

Why unit economics for verified AI requests beat consumption pricing — and how finance and engineering can align on a single metric.

THE REFRAME

Stop buying
tokens.
Start pricing
decisions.

280× cheaper

Fall in the cost of a standard model query, Nov 2022 → Oct 2024 — yet enterprise AI bills keep climbing.

\$644B

Forecast worldwide generative-AI spend in 2025 — up roughly 76% year over year.

01

THE CONSUMPTION
PRICING TRAP

*"Falling unit prices
and rising bills are
not a contradiction
— they are Jevons'
paradox arriving on
schedule."*

95%

of enterprise generative-AI
pilots show no measurable
P&L impact.⁴

WHY IT MATTERS

Token price is a procurement
metric. It cannot tell finance
whether a workload earns its
keep.

THE PROBLEM

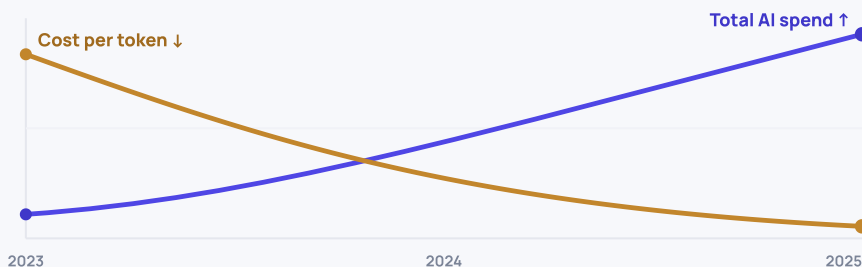
Cheaper tokens, bigger bills, weaker answers

Consumption pricing — paying per token or per call — won the early AI market because it is simple to meter and easy to start. As a **management** metric, it is failing. The price of intelligence is collapsing: the cost of a standard model query fell roughly **280-fold between late 2022 and late 2024**¹. Yet enterprise AI spending is accelerating, with worldwide generative-AI outlay forecast near **\$644 billion in 2025**².

This is not a paradox to explain away — it is **Jevons' paradox**: when a resource gets cheaper, total consumption rises rather than falls³. Lower unit prices invite more calls, longer context, more retries, and more agents. The bill grows even as each token gets cheaper, and the line item that finance sees has no relationship to the value the business received.

The Jevons gap — unit price down, total spend up

Indexed, directional (2023 = 100)



DIRECTIONAL ILLUSTRATION · SOURCES: STANFORD HAI AI INDEX 2025 [1]; GARTNER 2025 [2]

THREE WAYS CONSUMPTION PRICING MISLEADS

- 1 • It is unpredictable.** Bills scale with tokens, not outcomes, so a single prompt change or a chatty agent can move spend by double digits — making forecasting and budget ownership nearly impossible. Managing AI/ML spend is now a top FinOps priority for exactly this reason.⁶
- 2 • It carries no quality signal.** A correct answer and a confidently wrong one cost the same. Consumption metrics reward volume, not verified value — and over 40% of agentic projects are now forecast to be canceled by 2027 on escalating cost and unclear return.⁵
- 3 • It splits the organization.** Engineering optimizes latency and tokens; finance optimizes the invoice. With no shared unit of value, the two functions argue past each other.

02

A NEW UNIT
OF ACCOUNT

*"Price the decision
you can trust, not
the tokens it took to
reach it."*

DEFINITION

A **verified request** is one whose output has passed your acceptance test — schema, policy, evaluation, or human review — and is fit to act on.

THE SHIFT

From tokens consumed to decisions verified

The fix is not a cheaper token — it is a better unit of account. Move the metric from **what the model consumed** to **what the business can trust**: the verified request. The governing number becomes **Cost per Verified Request (CPVR)** — the fully-loaded cost of producing one output good enough to act on.

THE DECISION-ECONOMICS METRIC

$$\text{CPVR} = \frac{\text{Tokens} + \text{Retries} + \text{Verification} + \text{Human Review}}{\text{Verified, action-ready outputs}}$$

Every cost of getting to a trustworthy answer sits in the numerator; only outputs that pass your acceptance test count in the denominator. Failed and discarded generations raise the cost of the ones that survive.

THE VERIFICATION TAX

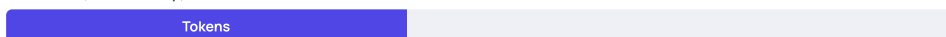
Headline token price describes a perfect world where every generation is usable. Real workloads aren't: a share of outputs fail validation, get retried, or need human correction. Those costs don't vanish — they load onto the outputs that succeed. We call this the **verification tax**, and it is invisible to consumption metrics.

Headline cost vs. fully-loaded cost per verified output

Illustrative workload at a 30% rework rate

Headline (tokens only)

1.0×



Fully-loaded per verified output

≈ 2.4×



■ Model tokens ■ Retries / regeneration ■ Automated verification ■ Human review

ILLUSTRATIVE · STRUCTURE, NOT A BENCHMARK

Once CPVR is in view, the optimization target changes. The cheapest path is no longer the fewest tokens — it is the **highest verified-yield per dollar**: better prompts and retrieval, right-sized models, and automated verification that catches failures before a human has to.

1.4–3×

Typical gap between headline token cost and fully-loaded cost per verified output once retries and review are counted.

03

BUILD THE
AI UNIT P&L

"If you can price one verified outcome, you can run AI like a business — not a science project."

THE ONE LINE THAT MATTERS

AI gross margin = value per verified outcome – fully-loaded cost per verified outcome.

CAUTION

Numbers shown are illustrative structure for a support-resolution agent — replace with your instrumented data.

IN PRACTICE

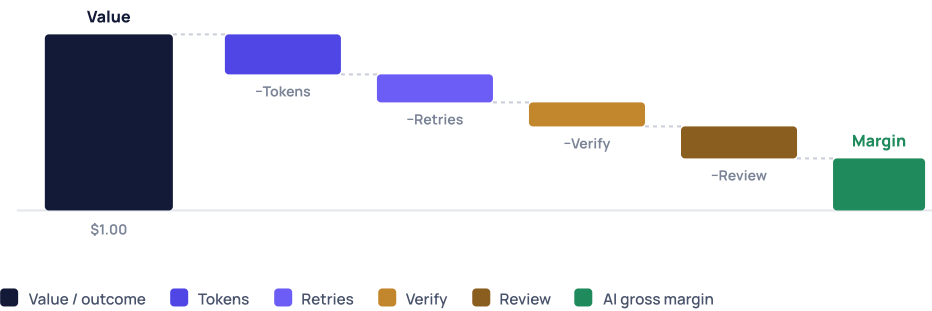
Run every workload as a unit P&L

Decision economics becomes operational when each workload carries its own miniature P&L: the value of one verified outcome, minus every cost incurred to produce it, equals the margin that workload returns. Below, the two lenses on the same workload — and the waterfall from value to margin.

DIMENSION	CONSUMPTION VIEW	DECISION-ECONOMICS VIEW
Unit of account	Tokens / API calls	Verified, action-ready outcomes
Predictability	Volatile; scales with usage	Stable; scales with outcomes
Quality signal	None — right and wrong cost alike	Built in — only passes count
Finance sees	An unpredictable invoice	Cost & margin per outcome
Engineering optimizes	Tokens & latency	Verified yield per dollar

AI unit P&L — from value to margin, per verified outcome

Illustrative - support-resolution agent



ILLUSTRATIVE STRUCTURE - REPLACE WITH INSTRUMENTED DATA

With this picture, decisions get sharp: workloads with healthy margin scale; thin-margin ones get re-engineered or retired. The same lens prices a resolved ticket, a merged pull request, or a processed claim — one comparable unit of economic truth across every AI investment.

04

ALIGN ON
ONE METRIC

"When finance and engineering share one number, the argument ends and the optimization begins."

90-DAY ADOPTION

- ① Instrument cost & verification per request
- ② Define the acceptance test per workload
- ③ Publish CPVR & margin on one dashboard
- ④ Set targets; review on a monthly cadence

THE OPERATING MODEL

One metric, two functions, no more arguments

CPVR works because it is legible to both sides. Engineering can move it with model choice, prompting, retrieval, and verification. Finance can forecast and budget on it because it scales with outcomes, not chatter. The same number sits on both dashboards — so the conversation shifts from “why is the bill so high?” to “which workloads earn the most verified value per dollar?”

WHO OPTIMIZES WHAT

ENGINEERING OWNS

Verified yield per dollar

Right-size models, cut retries with better prompting and retrieval, and automate verification so failures are caught before they reach a person.

Model routing

Eval gates

Retrieval

FINANCE OWNS

Margin & forecast

Budget per outcome, forecast on verified volume, and steer investment toward workloads with durable AI gross margin.

Unit forecast

Margin targets

Portfolio

The single decision-economics dashboard

One scorecard both functions read the same way

CPVR

Cost per verified request

Yield

% outputs verified first-pass

Margin

Value – loaded cost / outcome

Volume

Verified outcomes / period

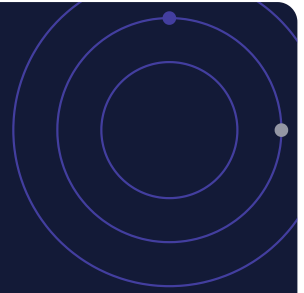
THE LEADERSHIP TAKEAWAY

The cost of intelligence will keep falling; your AI bill will keep rising. The enterprises that win won't be those who chase the cheapest token — they'll be those who can **price a verified decision** and run every workload to a margin. Make CPVR the number finance and engineering share, and AI stops being an unbounded cost and becomes a managed, compounding asset.

AI ENGINEERING STUDIO

We make AI work for your business — and your software.

O3Xs is an AI engineering studio that builds systems to engineer trust into how enterprises operate and deliver. We bring decision economics to the work itself — pricing verified outcomes, not raw consumption — with one team accountable for results.



THE O³ OPERATING MODEL

Diagnose before you automate, implement where ROI is provable, and own the outcome after go-live.

01

Optimize

We map value leakage across operations, revenue, and delivery — quantifying what should not be automated before any tool is selected.

PERFORMANCE DIAGNOSTIC

02

Orchestrate

We design and deploy AI workflows integrated with existing systems — no rip-and-replace, tied to business outcomes.

ROI-LINKED IMPLEMENTATION

03

Operate

We stay post go-live — monitoring KPIs, detecting drift, and iterating until results compound.

MANAGED OPERATIONS LAYER

SERVICE 01 · CONSULTATION

Business process automation

End-to-end automation of how your business already runs — mapping leakage, orchestrating AI across existing workflows, and operating until margin and performance compound.

Maximize results

Minimize waste

Verified impact

SERVICE 02 · FORGE AI

Enterprise SDLC platform

AI-native software delivery on your repository — intake through verification and merge-ready output — engineered to lower the cost per verified feature without replacing your stack.

Faster releases

Lower cost / feature

Governed scale

Put a price on your verified outcomes.

Book a diagnostic — a structured session to map where AI can recover value. Contact [\[email protected\]](#)

o3xs.com

SOURCES

References

This briefing synthesizes publicly available research from leading analyst and research institutions. Figures are drawn from the most recent editions available at time of writing (April 2026) and are cited for directional benchmarking; illustrative figures are labeled as such.

[1]

Stanford HAI. "Artificial Intelligence Index Report 2025" – sharp decline in inference cost; the cost of a standard model query fell by roughly two orders of magnitude ($\approx 280\times$) between late 2022 and late 2024. *Stanford Institute for Human-Centered AI, 2025.*

[2]

Gartner. Worldwide generative-AI spending forecast to total approximately \$644 billion in 2025, up roughly 76% year over year. *Gartner, Inc., 2025.*

[3]

Jevons' Paradox. Efficiency gains that lower the unit cost of a resource tend to increase, not decrease, its total consumption – invoked for AI inference by industry leadership in 2025. *Economic principle; contemporary commentary, 2025.*

[4]

MIT (Project NANDA). "The GenAI Divide: State of AI in Business 2025" – approximately 95% of enterprise generative-AI pilots fail to deliver measurable bottom-line impact. *Massachusetts Institute of Technology, 2025.*

[5]

Gartner. "Over 40% of Agentic AI Projects Will Be Canceled by End of 2027" – escalating costs and unclear business value among the leading causes. *Gartner, Inc., 2025.*

[6]

FinOps Foundation. "State of FinOps 2025" – managing AI/ML and cloud spend ranked among the top practitioner priorities, formalized in the "FinOps for AI" scope. *FinOps Foundation, 2025.*

[7]

O3Xs. AI Engineering Studio – O3 Consultation Framework and Forge AI overview. *o3xs.com, 2026.*

DISCLAIMER

This document is provided by O3Xs for general information and discussion purposes only and does not constitute professional, legal, financial, or investment advice. Third-party statistics are attributed to their respective owners and reproduced for commentary and benchmarking; figures marked "illustrative" depict structure, not measured results. O3Xs makes no representation as to completeness or accuracy and accepts no liability for decisions taken in reliance on this material. All trademarks and research are the property of their respective owners.

CONFIDENTIALITY & CONTACT

Confidential. This report is intended solely for the named recipient and authorized colleagues. It may not be copied, distributed, or disclosed to any third party, in whole or in part, without the prior written consent of O3Xs.

Contact. O3Xs – AI Engineering Studio · [\[email protected\]](#) · o3xs.com
Enterprise AI Readiness Series · O3Xs Research · April 2026 · v1.0